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Hello,

The following submission has been created.

Track Name: AIC2026

Paper ID: 2621

Paper Title: Enhancing Ad Performance: Automated Copy Generation and Sentiment Analysis Using Transformer Models

Abstract:

Background. The performance-marketing landscape has shifted toward extreme personalization and rapid creative refresh, a tempo that traditional manual copywriting can no longer sustain. Because platforms such as Google Ads and Meta Ads optimize delivery against engagement signals, advertisers must continually supply fresh, emotionally aligned variants. This creates a clear need for systems that can both produce ad copy at volume and forecast its likely emotional reception.

Objectives. This study designs and validates an end-to-end pipeline that produces high-conversion ad copy automatically. Specifically, we fine-tune a Large Language Model (LLM) to generate context-aware marketing text and then route every candidate through a sentiment-analysis stage that filters and refines the output. The intent is to lower creative fatigue while guaranteeing that every deployed variant clears a defined positive-sentiment threshold.

Methodology. The pipeline is implemented in Python on top of the Hugging Face Transformers ecosystem. A pre-trained GPT-2 checkpoint is fine-tuned on a curated corpus of high-performing ad headlines to emit diverse copy variants. A DistilBERT classifier then scores each variant; only those whose positive-class probability exceeds 0.85 are retained. The approach blends NLP techniques with practical performance-marketing heuristics, narrowing the gap between creative judgement and algorithmic rigor.

Results. Initial experiments show roughly a 40% reduction in the creative-development cycle time. The sentiment gate consistently flagged and discarded low-engagement copy, leaving 100% of approved variants aligned with brand-positive messaging. Comparative review further suggests that AI-assisted copy preserves semantic coherence on par with human writing while offering substantially better scalability for A/B testing.